

Day 3_Interview Questions

Q1. System role vs instructions in the user message?

System sets persistent behavior for the whole chat and carries higher priority; a user message is one turn's request. System = "who you are for this chat," user = "what I want now." Trap: they're not the same — order and persistence differ.

Q2. Why must the system message go first?

The model reads in order. If "I love you" is processed before "you are my manager," the persona isn't set yet and context breaks. System first = rules before requests.

Q3. What does temperature do under the hood?

The model outputs a probability distribution over the next token; temperature reshapes it before sampling. Low temp sharpens it (likely token dominates → safe). High temp flattens it (unlikely tokens get a chance → creative). That's the "why" behind the food-brand table.

Q4. Is temperature 0 fully deterministic?

Mostly, but not guaranteed. At 0 it picks the highest-probability token, so output is near-deterministic — but floating-point and backend batching can cause rare variation. Safe answer: "most deterministic, but not a hard guarantee."

Q5. LLM is a prediction machine, not a truth machine — explain.

It predicts the next likely token from training patterns; it doesn't look anything up. That's why it can be fluent but confidently wrong (hallucinate). Google retrieves facts; an LLM generates what sounds statistically right. (This sets up RAG later.)

Q6. Temperature use-case per task?

Doctor / legal / code-fix agent → ~0 (facts, consistency). Story / brand-name / marketing → high, 1–2 (novelty). Matching temp to task is the point.

Q7. Valid range and default?

[0, 2]; passing 3 errors. Default is model-dependent (often ~0.7–1) — don't say "default is always 0."

Q8. Related parameters?

top_p (nucleus sampling), max_tokens (output cap), n (number of answers), stop (halt sequences), frequency/presence_penalty (reduce repetition). Trap: "temperature vs top_p?" → both control randomness; tune one, not both.

Traps

- System role $\neq$ guaranteed obedience — a strong user prompt can override a weak system prompt (ties to prompt injection).
- High temp $\neq$ smarter — more random, not more correct, often less reliable.
- “How to make output reproducible?” $\rightarrow$ temperature 0 + a fixed seed (if supported).